

Three-Regime Convergence of the Snow Distance

Unified Proof under the Per-Step (Model A) Random-Step Variant

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Abstract

asdf

Contents

1	Setup	1
1.1	Notation (recalled from [1])	2
1.2	Four candidate snowfall algorithms (recalled from [1])	3
2	Three-regime classification	4
3	Theorem A	4
3.1	Centred height: two preparatory facts	4
3.2	Single-step Φ drift	5
3.3	Shape decomposition	6
3.4	Foster's drift criterion and moment bound	8
3.5	Doeblin minorization	11
3.6	Theorem A: statement and proof	11
4	Theorem B	12
5	Theorem C: Drift regime	14

1 Setup

This section is recall from [1]: graph/state/distance notation (§1.1) and the four candidate snowfall algorithms A, B, C, D that share that notation but differ in the increment-sampling rule (§1.2). The

present document analyses **Algorithm A**; the others are listed for reference and comparison. New notation introduced specifically for this document (state-function variants of h, Φ, M , the bilinear functional $B(S)$, etc.) is deferred to §3.6.

1.1 Notation (recalled from [1])

Graph and parameters. Let $\mathcal{G} = (V, \mathbf{E}, \mathbf{W})$ be a connected weighted graph with vertex set $V = \{v_1, \dots, v_n\}$, $|V| = n < \infty$, symmetric non-negative edge weights $\mathbf{W} = [w_{ij}]$, and $w_{ii} = 0$. Fix:

- $b > 0$: snowfall scale (per-step increment ceiling),
- $T_{\max} \in \mathbb{N}$: maximum consecutive flow-step count,
- μ_0 : a probability measure on V with $\mu_{\min} := \min_i \mu_0(v_i) > 0$ (full support).

A typical choice is $\mu_0 \propto \deg$ (degree-proportional) or uniform.

State variables. At each integer time $t \geq 0$, the algorithm maintains:

- $h(v_i, t) \in \mathbb{R}$ for each $v_i \in V$: the snow *height* at v_i ;
- $X_t \in V$: the *walker* location;
- $Z_t \in \{0, 1, \dots, T_{\max}\}$: the *block timer* (consecutive flow-step count).

The initial height is fixed at the zero signal $\mathbf{y} = 0$ throughout this document: $h(v_i, 0) = 0$ for all v_i . Consequently $\bar{h}(v_i, 0) = 0$ and $M(0) = \Phi(0) = 0$. The initial walker $X_0 \sim \mu_0$ and $Z_0 := T_{\max}$ (so that step 1 starts with a fresh fall).

Augmented state. The augmented Markov chain underlying the analysis is

$$S_t := (\delta_t, X_t, Z_t) \in \mathcal{X}^* := \mathbb{R}^{n-1} \times V \times \{0, 1, \dots, T_{\max}\},$$

where $\delta_t := (h(v_2, t) - h(v_1, t), \dots, h(v_n, t) - h(v_1, t)) \in \mathbb{R}^{n-1}$ is the height-difference vector (tracking only relative heights, since absolute height grows without bound). The natural filtration is $\mathcal{F}_t := \sigma(X_0, b'_1, X_1, \dots, b'_t, X_t)$.

Snow distance. The quantity of interest is the cumulative squared height difference between two nodes:

$$SD_{ij}^2(t) := \sum_{s=0}^t (h(v_i, s) - h(v_j, s))^2.$$

Long-run deposition rate. Whenever the limit exists, the *long-run deposition rate* at node v_i is

$$\rho_i := \lim_{t \rightarrow \infty} \frac{1}{t} \sum_{s=1}^t \mathbf{1}\{X_s = v_i\}.$$

The value of ρ_i depends on μ_0 , the graph topology, and the flow/block dynamics. Existence of ρ_i is established unconditionally in [2] on every connected graph with $\mu_{\min} > 0$; see Appendix ??.

Centred height and Lyapunov function.

$$\bar{h}(t) := \frac{1}{n} \sum_{i=1}^n h(v_i, t), \quad \bar{h}(v_i, t) := h(v_i, t) - \bar{h}(t), \quad \Phi(t) := \sum_{i=1}^n \bar{h}(v_i, t)^2.$$

$M(t) := \max_i |\bar{h}(v_i, t)|$ denotes the (one-sided) centred height range; equivalently, by Fact 2 below, $\max_i \bar{h}(v_i, t) - \min_i \bar{h}(v_i, t) \in [M(t), 2M(t)]$.

1.2 Four candidate snowfall algorithms (recalled from [1])

The Heavy-Snow Transform of [1] fixes the walker rule (block / flow) but leaves the per-step increment b' open. Four candidate models have been considered in the Snowman Lab paper team:

Model	Increment draw	Per-step increment
Algorithm A (per-step i.i.d.)	each step	$b'_s \sim \text{Unif}(0, b)$, i.i.d. across s
Algorithm B (per-round constant)	each round r	$b^{(r)} \sim \text{Unif}(0, b)$; used for every step of round r
Algorithm C (per-round with decay)	each round r	$b^{(r)} f(s)$ for some decay function f
Algorithm D (deterministic)	—	$b'_s = b$ (constant)

Algorithm A (the model of this document). At each step $t + 1$ ($t \geq 0$):

1. **Draw increment.** Sample $b'_{t+1} \sim \text{Unif}(0, b)$ independently of all past randomness.
2. **Choose walker.** If $Z_t = T_{\max}$ (block timer expired), set $X_{t+1} \sim \mu_0$ (a *block* step: fresh fall). Otherwise apply the *flow rule*: among the neighbours of X_t in $\mathcal{G}$, restrict to those with height $\leq h(X_t, t)$ (the *downstream set* $\mathcal{D}_t$); if $\mathcal{D}_t = \emptyset$, the flow stalls and $X_{t+1} \sim \mu_0$ as in a block step. Otherwise X_{t+1} is chosen from $\mathcal{D}_t$ by degree-weighted probability.
3. **Update height.** $h(v_i, t + 1) = h(v_i, t) + b'_{t+1} \mathbf{1}\{v_i = X_{t+1}\}$ for every $v_i \in V$.
4. **Update timer.** $Z_{t+1} = 0$ if step $t + 1$ was a block step; $Z_{t+1} = Z_t + 1$ otherwise. We say t_r is a *block-flag time* when $Z_{t_r} = T_{\max}$; a *round* is the segment $(t_r, t_{r+1}]$ of length $m_r := t_{r+1} - t_r \in [1, T_{\max} + 1]$.

Algorithms B, C, D (reference only). The other models share the walker rule above and differ only in step (1):

- **Algorithm B** (per-round constant): at the start of each round r , draw $b^{(r)} \sim \text{Unif}(0, b)$ once and use $b'_t := b^{(r)}$ for every step t in that round. Round total $= m_r \cdot b^{(r)}$.
- **Algorithm C** (per-round with decay): same per-round draw $b^{(r)} \sim \text{Unif}(0, b)$, but the increment decays within the round, $b'_{t_r+s} := b^{(r)} f(s)$ for a decay function $f : \{0, \dots, m_r - 1\} \rightarrow [0, 1]$ (e.g. $f(s) = r^s$ exponential or $f(s) = (1 - s/L)_+$ linear).
- **Algorithm D** (deterministic): $b'_s = b$ for every step. No randomness in the increment.

All four share the marginal first moment $\mathbb{E}[b'_s] = b/2$ for A–C and $b'_s = b$ for D; the second moment is $\mathbb{E}[(b'_s)^2] = b^2/3$ for A–C and b^2 for D.

Status note. The Snowman Lab paper team has officially adopted Algorithm B as of 2026-05-17 (chair’s decision). The present document analyses Algorithm A as the reference baseline; a companion file re-derives the leading constants under Algorithm B and shows that the leading constants of Theorems B and C are preserved, while the Algorithm B counterpart of c_{ij} (Theorem A) may shift at the graph-specific numerical level. Algorithms C and D are listed for completeness; Algorithm D is not adopted because it fails the lattice-irreducibility argument that motivates the random-step variant in the first place.

2 Three-regime classification

Snow distance and long-run deposition rates ρ_i were defined in Section 1. We now classify pairs (i, j) by these rates.

Definition 1 (Regime classification). *Suppose ρ_i exists (a.s.) for all $i \in V$.*

- **Balanced regime:** $\rho_i = 1/n$ for all i (since $\sum_i \rho_i = 1$, this is equivalent to $\rho_i = \rho_j$ for all i, j and global balance).
- **Drift regime (pair i, j):** $\rho_i \neq \rho_j$.
- **Intermediate regime (pair i, j):** $\rho_i = \rho_j$ but the system is globally unbalanced (there exist k, ℓ with $\rho_k \neq \rho_\ell$).

3 Theorem A

This section develops the lemmas needed for Theorem 1. All proofs are for Model A of the random-step variant: increments $b'_t \sim \text{Unif}(0, b)$ drawn i.i.d. at every step, independent of all other randomness. The corresponding moments

$$\bar{b} := \mathbb{E}[b'_t] = b/2, \quad \mathbb{E}[(b'_t)^2] = b^2/3, \quad \text{Var}(b'_t) = b^2/12$$

are used throughout.

3.1 Centred height: two preparatory facts

Define $\bar{h}(v_i, t) := h(v_i, t) - \bar{h}(t)$ (centred height) and $\Phi(t) := \sum_i \bar{h}(v_i, t)^2$.

Fact 1 (single-step update of $\bar{h}(v_i, t)$). *Fix $t \geq 0$. Suppose at step $t + 1$ the increment $b'_{t+1} \sim \text{Unif}(0, b)$ is deposited at the walker location $X_{t+1} \in V$. Then for every node $v_i \in V$,*

$$\bar{h}(v_i, t + 1) = \begin{cases} \bar{h}(v_i, t) + b'_{t+1} \frac{n-1}{n}, & v_i = X_{t+1}, \\ \bar{h}(v_i, t) - \frac{b'_{t+1}}{n}, & v_i \neq X_{t+1}. \end{cases}$$

Proof. The height update is $h(v_i, t + 1) = h(v_i, t) + b'_{t+1} \mathbf{1}\{v_i = X_{t+1}\}$. Summing over i and dividing by n ,

$$\bar{h}(t + 1) = \frac{1}{n} \sum_{i=1}^n h(v_i, t + 1) = \bar{h}(t) + \frac{b'_{t+1}}{n}.$$

Substitute both updates into $\bar{h}(v_i, t+1) := h(v_i, t+1) - \bar{h}(t+1)$. For the recipient $v_i = X_{t+1}$:

$$\begin{aligned}\bar{h}(X_{t+1}, t+1) &= [h(X_{t+1}, t) + b'_{t+1}] - [\bar{h}(t) + \frac{b'_{t+1}}{n}] \\ &= \bar{h}(X_{t+1}, t) + b'_{t+1} \frac{n-1}{n}.\end{aligned}$$

For any other node $v_i \neq X_{t+1}$:

$$\begin{aligned}\bar{h}(v_i, t+1) &= h(v_i, t) - [\bar{h}(t) + \frac{b'_{t+1}}{n}] \\ &= \bar{h}(v_i, t) - \frac{b'_{t+1}}{n}.\end{aligned}$$

□

Fact 2 ($\sum_i \bar{h}(v_i, t) = 0$). *For every t ,*

$$\sum_{i=1}^n \bar{h}(v_i, t) = \sum_{i=1}^n (h(v_i, t) - \bar{h}(t)) = \sum_{i=1}^n h(v_i, t) - n \bar{h}(t) = 0,$$

since $\bar{h}(t) := \frac{1}{n} \sum_{i=1}^n h(v_i, t)$.

Fact 3 (Pigeonhole bounds on Φ). *For every t ,*

$$\frac{M(t)^2}{2} \leq \Phi(t) \leq n M(t)^2,$$

where $M(t) := \max_{v \in V} |\bar{h}(v, t)|$.

$$\begin{aligned}\Phi(t) &= \sum_i \bar{h}(v_i, t)^2 \geq \bar{h}_{\max}^2 + \bar{h}_{\min}^2 \\ &\geq \frac{(\bar{h}_{\max} + |\bar{h}_{\min}|)^2}{2} = \frac{M(t)^2}{2} \quad (\because (A^2 + B^2) \geq (A+B)^2/2, \sum_i \bar{h}(v_i, t) = 0) \\ \Phi(t) &\leq n M(t)^2 \quad (\because n \text{ terms each } \leq M(t)^2).\end{aligned}$$

3.2 Single-step Φ drift

Lemma 1 (Single-step Φ drift identity). *For any $t \geq 0$ and $X_{t+1} \in V$,*

$$\mathbb{E}[\Phi(t+1) - \Phi(t) \mid \mathcal{F}_t, X_{t+1}] = b \bar{h}(X_{t+1}, t) + \frac{(n-1)b^2}{3n}.$$

Proof. We compute $\Phi(t+1) - \Phi(t)$ in closed form, then take expectation over $b'_{t+1} \sim \text{Unif}(0, b)$.

Split $\Phi(t)$ by the recipient node and substitute Fact 1 into $\Phi(t+1)$:

$$\begin{aligned}\Phi(t) &= \bar{h}(X_{t+1}, t)^2 + \sum_{u \neq X_{t+1}} \bar{h}(u, t)^2, \\ \Phi(t+1) &= [\bar{h}(X_{t+1}, t) + b'_{t+1} \frac{n-1}{n}]^2 + \sum_{u \neq X_{t+1}} \left[\bar{h}(u, t) - \frac{b'_{t+1}}{n} \right]^2 \quad (\because \text{Fact 1}).\end{aligned}$$

Split the difference into two terms by recipient: $\Phi(t+1) - \Phi(t) = \text{Term}_1 + \text{Term}_2$, where

$$\begin{aligned}\text{Term}_1 &:= [\bar{h}(X_{t+1}, t) + b'_{t+1} \frac{n-1}{n}]^2 - \bar{h}(X_{t+1}, t)^2, \\ \text{Term}_2 &:= \sum_{u \neq X_{t+1}} \left\{ \left[\bar{h}(u, t) - \frac{b'_{t+1}}{n} \right]^2 - \bar{h}(u, t)^2 \right\}.\end{aligned}$$

Compute Term_1 using $(A + B)^2 - A^2 = 2AB + B^2$ with $A = \hbar(X_{t+1}, t)$ and $B = b'_{t+1}(n - 1)/n$:

$$\text{Term}_1 = 2\hbar(X_{t+1}, t) \cdot \frac{(n-1)b'_{t+1}}{n} + \frac{(n-1)^2(b'_{t+1})^2}{n^2}.$$

Compute Term_2 termwise (each summand uses $(A + B)^2 - A^2$ with $B = -b'_{t+1}/n$), then collapse using Fact 2:

$$\begin{aligned} \text{Term}_2 &= \sum_{u \neq X_{t+1}} \left[-\frac{2b'_{t+1}}{n} \hbar(u, t) + \frac{(b'_{t+1})^2}{n^2} \right] && (\because (A + B)^2 - A^2 = 2AB + B^2) \\ &= -\frac{2b'_{t+1}}{n} \sum_{u \neq X_{t+1}} \hbar(u, t) + (n - 1) \cdot \frac{(b'_{t+1})^2}{n^2} && (\because n - 1 \text{ summands of } (b'_{t+1})^2/n^2) \\ &= -\frac{2b'_{t+1}}{n} \cdot (-\hbar(X_{t+1}, t)) + \frac{(n-1)(b'_{t+1})^2}{n^2} && (\because \text{Fact 2}) \\ &= \frac{2b'_{t+1}}{n} \hbar(X_{t+1}, t) + \frac{(n-1)(b'_{t+1})^2}{n^2}. \end{aligned}$$

Combine $\text{Term}_1 + \text{Term}_2$ by collecting common factors of $2b'_{t+1} \hbar(X_{t+1}, t)$ and $(b'_{t+1})^2$:

$$\begin{aligned} \Phi(t + 1) - \Phi(t) &= \text{Term}_1 + \text{Term}_2 \\ &= 2\hbar(X_{t+1}, t) \cdot \frac{(n-1)b'_{t+1}}{n} + \frac{(n-1)^2(b'_{t+1})^2}{n^2} + \frac{2b'_{t+1}}{n} \hbar(X_{t+1}, t) + \frac{(n-1)(b'_{t+1})^2}{n^2} \\ &= 2b'_{t+1} \hbar(X_{t+1}, t) \cdot \underbrace{\left[\frac{n-1}{n} + \frac{1}{n} \right]}_{=1} + (b'_{t+1})^2 \cdot \underbrace{\frac{(n-1)^2 + (n-1)}{n^2}}_{=(n-1)/n} \\ &= 2b'_{t+1} \hbar(X_{t+1}, t) + \frac{n-1}{n} (b'_{t+1})^2. \end{aligned}$$

Take expectation over $b'_{t+1} \sim \text{Unif}(0, b)$ using $\mathbb{E}[b'_{t+1}] = b/2$ and $\mathbb{E}[(b'_{t+1})^2] = b^2/3$:

$$\begin{aligned} \mathbb{E}[\Phi(t + 1) - \Phi(t) \mid \mathcal{F}_t, X_{t+1}] &= 2 \cdot \frac{b}{2} \cdot \hbar(X_{t+1}, t) + \frac{n-1}{n} \cdot \frac{b^2}{3} \\ &= b \hbar(X_{t+1}, t) + \frac{(n-1)b^2}{3n}. \end{aligned}$$

□

3.3 Shape decomposition

Let t_r be the r -th block-flag time, $m_r := t_{r+1} - t_r$ the round length, and $\tilde{X}_j := X_{t_r+j}$. Let $\#_u := |\{j : \tilde{X}_j = u\}|$ be the visit count and $\pi(u, S) := \mathbb{E}[\#_u \mid S_{t_r}]$ the occupation measure.

Lemma 2 (Shape decomposition).

$$\begin{aligned} \sum_{j=1}^{m_r} \hbar(\tilde{X}_j, t_r + j - 1) &= \sum_{u \in V} \#_u \hbar(u, t_r) + \Delta\text{Shape}^{(r)}, \\ |\Delta\text{Shape}^{(r)}| &\leq C_{\Delta\text{Shape}} := \frac{(T_{\max}+1)T_{\max}}{2} b. \end{aligned}$$

Proof. Apply Fact 1 at each of the $j - 1$ intermediate deposits to expand $\hbar(\tilde{X}_j, t_r + j - 1)$ around its round-start value $\hbar(\tilde{X}_j, t_r)$:

$$\hbar(\tilde{X}_j, t_r + j - 1) = \hbar(\tilde{X}_j, t_r) + \sum_{k=1}^{j-1} b'_{t_r+k} \left[\mathbf{1}\{\tilde{X}_k = \tilde{X}_j\} - \frac{1}{n} \right].$$

Sum over $j = 1, \dots, m_r$ and split by the two summands:

$$\sum_{j=1}^{m_r} \hbar(\tilde{X}_j, t_r + j - 1) = \underbrace{\sum_{j=1}^{m_r} \hbar(\tilde{X}_j, t_r)}_{=: \text{Term}_1} + \underbrace{\sum_{j=1}^{m_r} \sum_{k=1}^{j-1} b'_{t_r+k} \left[\mathbf{1}\{\tilde{X}_k = \tilde{X}_j\} - \frac{1}{n} \right]}_{=: \text{Term}_2 =: \Delta\text{Shape}^{(r)}}.$$

Compute Term_1 by re-indexing the sum over j via the visit count $\#_u := |\{j : \tilde{X}_j = u\}|$:

$$\text{Term}_1 = \sum_{j=1}^{m_r} \hbar(\tilde{X}_j, t_r) = \sum_{u \in V} \#_u \hbar(u, t_r).$$

Bound on $\Delta\text{Shape}^{(r)} := \text{Term}_2$: each summand satisfies

$$\begin{aligned} \left| b'_{t_r+k} \left[\mathbf{1}\{\tilde{X}_k = \tilde{X}_j\} - \frac{1}{n} \right] \right| &\leq b'_{t_r+k} \cdot \max\left\{1 - \frac{1}{n}, \frac{1}{n}\right\} \quad (\because |x - \frac{1}{n}| \leq \max(1 - \frac{1}{n}, \frac{1}{n}) \text{ for } x \in \{0, 1\}) \\ &\leq b \cdot \frac{n-1}{n} \quad (\because b'_{t_r+k} \leq b) \\ &\leq b. \end{aligned}$$

Count the number of (j, k) pairs in the double sum defining $\Delta\text{Shape}^{(r)}$:

$$\sum_{j=2}^{m_r} \sum_{k=1}^{j-1} 1 = \sum_{j=2}^{m_r} (j-1) = \frac{m_r(m_r-1)}{2} \leq \frac{(T_{\max}+1)T_{\max}}{2} \quad (\because m_r \leq T_{\max} + 1).$$

Combining the per-summand bound and the pair count,

$$|\Delta\text{Shape}^{(r)}| \leq b \cdot \frac{(T_{\max}+1)T_{\max}}{2} = C_{\Delta\text{Shape}}.$$

□

Lemma 3 (Round-level Φ drift). *Define $\pi(u, S) := \mathbb{E}[\#_u \mid S_{t_r}]$ and $B(S) := \sum_{u \in V} \pi(u, S) \hbar(u, t_r)$. For every round $r \geq 0$, the round-level Φ drift decomposes as*

$$\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] = b B(S_{t_r}) + C_{\text{rem}}^{(r)},$$

where $C_{\text{rem}}^{(r)}$ is a remainder term (a function of S_{t_r}) satisfying the universal bound

$$|C_{\text{rem}}^{(r)}| \leq C_{M\text{-free}} := (T_{\max} + 1) b^2 \left[\frac{T_{\max}}{2} + \frac{n-1}{3n} \right].$$

The constant $C_{M\text{-free}}$ depends only on $(n, b, T_{\max})$.

Proof.

$$\begin{aligned} \mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] &= \sum_{j=1}^{m_r} \mathbb{E} \left[b \hbar(\tilde{X}_j, t_r + j - 1) + \frac{(n-1)b^2}{3n} \mid S_{t_r} \right] \quad (\because \text{Lem. 1}) \\ &= b \mathbb{E} \left[\sum_{j=1}^{m_r} \hbar(\tilde{X}_j, t_r + j - 1) \mid S_{t_r} \right] + \frac{(n-1)b^2}{3n} \mathbb{E}[m_r \mid S_{t_r}] \\ &= b \cdot \text{Term}_1 + \text{Term}_2, \end{aligned}$$

where

$$\begin{aligned}\text{Term}_1 &:= \mathbb{E} \left[\sum_{j=1}^{m_r} \tilde{h}(\tilde{X}_j, t_r + j - 1) \mid S_{t_r} \right], \\ \text{Term}_2 &:= \frac{(n-1)b^2}{3n} \mathbb{E}[m_r \mid S_{t_r}].\end{aligned}$$

Compute Term_1 by substituting Lemma 2:

$$\begin{aligned}\text{Term}_1 &= \mathbb{E} \left[\sum_{u \in V} \#_u \tilde{h}(u, t_r) + \Delta \text{Shape}^{(r)} \mid S_{t_r} \right] \\ &= \sum_{u \in V} \pi(u, S) \tilde{h}(u, t_r) + \mathbb{E}[\Delta \text{Shape}^{(r)} \mid S_{t_r}] \\ &= B(S_{t_r}) + \mathbb{E}[\Delta \text{Shape}^{(r)} \mid S_{t_r}].\end{aligned}$$

Define the remainder

$$C_{\text{rem}}^{(r)} := b \mathbb{E}[\Delta \text{Shape}^{(r)} \mid S_{t_r}] + \frac{(n-1)b^2}{3n} \mathbb{E}[m_r \mid S_{t_r}].$$

Then $\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] = b B(S_{t_r}) + C_{\text{rem}}^{(r)}$ by the Term_1 computation above. Bound $|C_{\text{rem}}^{(r)}|$:

$$\begin{aligned}|C_{\text{rem}}^{(r)}| &\leq b |\mathbb{E}[\Delta \text{Shape}^{(r)} \mid S_{t_r}]| + \frac{(n-1)b^2}{3n} \mathbb{E}[m_r \mid S_{t_r}] && (\because |a+b| \leq |a| + |b|, m_r \geq 0) \\ &\leq b C_{\Delta \text{Shape}} + \frac{(T_{\max}+1)(n-1)b^2}{3n} && (\because \text{Lem. 2}, m_r \leq T_{\max} + 1) \\ &= (T_{\max} + 1) b^2 \left[\frac{T_{\max}}{2} + \frac{n-1}{3n} \right] && (\because C_{\Delta \text{Shape}} = (T_{\max} + 1) T_{\max} b/2) \\ &=: C_{M\text{-free}}.\end{aligned}$$

□

3.4 Foster's drift criterion and moment bound

For Foster–Lyapunov to apply (the Theorem A proof in §3.6), the round-level $\Phi(t_r)$ drift must be *strictly negative when $M(t_r)$ is large* — otherwise the height range $M(t_r)$ escapes to infinity and Harris recurrence cannot follow. The required condition is:

Assumption 1 (Foster's drift criterion). *There exist graph-level constants $\varepsilon_B > 0$ and $C_B \geq 0$ such that $B(S_{t_r})$, the expected weighted height deposit defined in Lemma 3, satisfies*

$$B(S_{t_r}) \leq -\varepsilon_B M(S_{t_r}) + C_B \quad \forall r \geq 0.$$

Fact 4 (Φ -drift bound from Assumption 1). *Composing Assumption 1 with Lemma 3 step by step:*

$$\begin{aligned}\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] &= b B(S_{t_r}) + C_{\text{rem}}^{(r)} && (\because \text{Lem. 3}) \\ &\leq b(-\varepsilon_B M(S_{t_r}) + C_B) + C_{\text{rem}}^{(r)} && (\because \text{Assumption 1: } B(S_{t_r}) \leq -\varepsilon_B M(S_{t_r}) + C_B) \\ &\leq b(-\varepsilon_B M(S_{t_r}) + C_B) + C_{M\text{-free}} && (\because |C_{\text{rem}}^{(r)}| \leq C_{M\text{-free}}) \\ &= -b \varepsilon_B M(S_{t_r}) + b C_B + C_{M\text{-free}} \\ &=: -\varepsilon_F M(S_{t_r}) + C_F && (\varepsilon_F := b \varepsilon_B, C_F := b C_B + C_{M\text{-free}})\end{aligned}$$

for all $r \geq 0$.

From Foster drift to moment bound. Fact 4 gives $\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] \leq -\varepsilon_F M(S_{t_r}) + C_F$, which is $M(S_{t_r})$ -linear. Converting to a $\Phi(t_r)$ -side drift via Fact 3 ($\Phi(t_r) \leq n M(t_r)^2 \Leftrightarrow M(t_r) \geq \sqrt{\Phi(t_r)/n}$),

$$\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] \leq -\varepsilon_F \sqrt{\Phi(t_r)/n} + C_F \quad (\text{of order } -\Phi(t_r)^{1/2}).$$

Both Φ and M are state functions $\mathcal{X}^* \rightarrow \mathbb{R}_{\geq 0}$ (via $\bar{h}(v, S) := h(v, S) - \bar{h}(S)$, $\Phi(S) := \sum_i \bar{h}(v_i, S)^2$, $M(S) := \max_{v \in V} |\bar{h}(v, S)|$, with $\Phi(t_r) \equiv \Phi(S_{t_r})$ used interchangeably). The round-skeleton chain $\{S_{t_r}\}_{r \geq 0}$ is ψ -irreducible and aperiodic on $\mathcal{X}^*$ (consequence of the Doeblin minorization, Proposition 2 below).

Lemma 4 (Foster–Lyapunov moment bound via Meyn–Tweedie). *Suppose the round-skeleton chain $\{S_{t_r}\}$ is ψ -irreducible and aperiodic on $\mathcal{X}^*$. Suppose there exist a Lyapunov function $V_{\text{MT}} : \mathcal{X}^* \rightarrow \mathbb{R}_{\geq 0}$, a state function $f_{\text{MT}} : \mathcal{X}^* \rightarrow [1, \infty)$ (outside a set C), a petite set $C \subset \mathcal{X}^*$, and a constant $b \geq 0$ such that*

$$\mathbb{E}[V_{\text{MT}}(S_{t_{r+1}}) - V_{\text{MT}}(S_{t_r}) \mid S_{t_r}] \leq -f_{\text{MT}}(S_{t_r}) + b \mathbf{1}_C(S_{t_r}) \quad \forall S_{t_r} \in \mathcal{X}^*.$$

Then there exists a unique invariant probability π^* on $\mathcal{X}^*$ with

$$\pi^*(f_{\text{MT}}) := \int_{\mathcal{X}^*} f_{\text{MT}}(S) \pi^*(dS) < \infty.$$

([4, Thm. 14.0.1], (iii) $\Rightarrow$ (i). Subscripts MT distinguish the Meyn–Tweedie statement variables from the vertex set V and the Algorithm-C decay f .)

Lemma 5 (A priori Φ -moment bound). *Under Assumption 1 and the Doeblin minorization (Proposition 2 below), the round-skeleton chain $\{S_{t_r}\}$ satisfies $\mathbb{E}_{\pi^*}[\Phi^r] < \infty$ for all $r < 1/2$.*

Proof. Take

$$\begin{aligned} V_{\text{MT}}(S) &:= \Phi(S), & f_{\text{MT}}(S) &:= \varepsilon_F \sqrt{\Phi(S)/n}, \\ C &:= \{S \in \mathcal{X}^* : \Phi(S) \leq n/\varepsilon_F^2\}, & b &:= C_F + 1. \end{aligned}$$

Step 1: $f_{\text{MT}} \geq 1$ outside C .

$$\begin{aligned} S \notin C &\implies \Phi(S) > n/\varepsilon_F^2 && (\because \text{definition of } C) \\ &\implies \sqrt{\Phi(S)/n} > 1/\varepsilon_F && (\because \sqrt{\cdot} \text{ monotone, divide by } n) \\ &\implies f_{\text{MT}}(S) = \varepsilon_F \sqrt{\Phi(S)/n} > 1 && (\because \text{multiply by } \varepsilon_F). \end{aligned}$$

Step 2: drift inequality matches Thm 14.0.1 (iii) form.

$$\begin{aligned} \mathbb{E}[V_{\text{MT}}(S_{t_{r+1}}) - V_{\text{MT}}(S_{t_r}) \mid S_{t_r}] &= \mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] && (\because V_{\text{MT}} = \Phi) \\ &\leq -\varepsilon_F M(S_{t_r}) + C_F && (\because \text{Fact 4}) \\ &\leq -\varepsilon_F \sqrt{\Phi(S_{t_r})/n} + C_F && (\because \text{Fact 3: } M \geq \sqrt{\Phi/n}) \\ &= -f_{\text{MT}}(S_{t_r}) + C_F && (\because \text{def. of } f_{\text{MT}}). \end{aligned}$$

Outside C , Step 1 gives $f_{\text{MT}} > 1$ so the RHS ≤ -1 . Inside C , the residual C_F is absorbed into $b \mathbf{1}_C$ with $b = C_F + 1$ ($\geq C_F + f_{\text{MT}}(S)$ on C , where $f_{\text{MT}} \leq 1$). Hence

$$\mathbb{E}[V_{\text{MT}}(S_{t_{r+1}}) - V_{\text{MT}}(S_{t_r}) \mid S_{t_r}] \leq -f_{\text{MT}}(S_{t_r}) + b \mathbf{1}_C(S_{t_r}) \quad \forall S_{t_r} \in \mathcal{X}^*.$$

C is a Φ -sub-level set, hence petite under the Doeblin minorization.

Step 3: apply Lemma 4.

$$\begin{aligned} \pi^*(f_{\text{MT}}) &= \mathbb{E}_{\pi^*}[\varepsilon_F \sqrt{\Phi/n}] < \infty & (\because \text{Lemma 4}) \\ \mathbb{E}_{\pi^*}[\sqrt{\Phi}] &< \infty & (\because \text{multiply by } \sqrt{n}/\varepsilon_F) \\ \mathbb{E}_{\pi^*}[\Phi^r] &< \infty \quad \forall r < 1/2 & (\because \text{Markov: } \Phi^r \leq \Phi^{1/2} + 1 \text{ for } r \leq 1/2). \end{aligned}$$

□

For Theorem A we need $r = 1$ (control $\mathbb{E}_{\pi^*}[(\delta_i - \delta_j)^2] \leq 2 \mathbb{E}_{\pi^*}[\Phi]$), so the bound of Lemma 5 is insufficient.

Second application $V_{\text{MT}} := V_{\text{Lifting}} = \Phi^2$: by the identity

$$V_{\text{Lifting}}(S_{t_{r+1}}) - V_{\text{Lifting}}(S_{t_r}) = 2 \Phi(t_r)(\Phi(t_{r+1}) - \Phi(t_r)) + (\Phi(t_{r+1}) - \Phi(t_r))^2,$$

the leading conditional term is of order $\Phi(t_r) \cdot M(t_r) \geq M(t_r)^3/\sqrt{2} \sim V_{\text{Lifting}}^{3/4}$ (via Fact 3; precise constant in Proposition 1 below). Setting

$$\begin{aligned} V_{\text{MT}}(S) &:= V_{\text{Lifting}}(S) = \Phi(S)^2, & f_{\text{MT}}(S) &:= V_{\text{Lifting}}(S)^{3/4} = \Phi(S)^{3/2}, \\ C' &:= \{S : V_{\text{Lifting}}(S) \leq 1\}, & f_{\text{MT}}|_{C'^c} &\geq 1, \end{aligned}$$

Lemma 4 yields

$$\mathbb{E}_{\pi^*}[\Phi^{3/2}] < \infty \implies \mathbb{E}_{\pi^*}[\Phi] < \infty \quad (\because \Phi \leq \Phi^{3/2} + 1, \text{ Markov}).$$

Proposition 1 (Moment bound). *Under the round drift (Lemma 3) and Doeblin minorization (Steps 1–2 of Theorem 1), $\mathbb{E}_{\pi^*}[\Phi] < \infty$.*

Proof. Set $V_{\text{Lifting}} := \Phi^2$ and $\Delta\Phi_r := \Phi(t_{r+1}) - \Phi(t_r)$.

Within-round height control. During a round of $m_r \leq T_{\text{max}} + 1$ steps, each step deposits $b' \leq b$ at one node, changing $\hbar$ by at most $b(n-1)/n < b$. Since $|\hbar(v, t_r)| \leq M(t_r)$ and each step perturbs $\hbar$ by $< b$, we have $|\hbar(v, t_r + j)| \leq M(t_r) + T_{\text{max}}b$ for all v, j within the round. The single-step Φ -change satisfies $|2b'\hbar + (n-1)(b')^2/n| \leq 2b(M + T_{\text{max}}b) + (n-1)b^2/n$. Summing over m_r steps:

$$|\Delta\Phi_r| \leq A_1 M + A_2 \quad \text{a.s.}, \quad A_1 := 2b(T_{\text{max}} + 1), \quad A_2 := (T_{\text{max}} + 1)[2bT_{\text{max}}b + (n-1)b^2/n]. \quad (1)$$

Drift of V_{Lifting} .

$$\begin{aligned} \mathbb{E}[\Delta V_{\text{Lifting}} \mid S_{t_r}] &= 2\Phi \mathbb{E}[\Delta\Phi_r \mid S] + \mathbb{E}[(\Delta\Phi_r)^2 \mid S] \\ &\leq 2\Phi(-\varepsilon_F M + C_r) + (A_1 M + A_2)^2. \end{aligned}$$

Pigeonhole: $\sum h_i = 0$ with range M forces $\Phi = \sum h_i^2 \geq M^2/2$ (since $h_{\max}^2 + h_{\min}^2 \geq (h_{\max} + |h_{\min}|)^2/2 = M^2/2$). Also $\Phi \leq nM^2$. Hence $-2\varepsilon_F \Phi M \leq -\varepsilon_F M^3$ and $2C_r \Phi \leq 2nC_r M^2$. Combining with (1):

$$\mathbb{E}[\Delta V_{\text{Lifting}} \mid S] \leq -\varepsilon_F M^3 + C_4 M^2 + C_5.$$

For $M \geq M_0 := 2C_4/(\varepsilon_F)$, the cubic term dominates: $\mathbb{E}[\Delta V_{\text{Lifting}} \mid S] \leq -\frac{\varepsilon_F}{2} M^3 + C_5$. From $\Phi \leq nM^2$: $M \geq (\Phi/n)^{1/2}$, so $M^3 \geq \Phi^{3/2}/n^{3/2} = V_{\text{Lifting}}^{3/4}/n^{3/2}$:

$$\mathbb{E}[\Delta V_{\text{Lifting}} \mid S] \leq -\frac{\varepsilon_F}{2n^{3/2}} V_{\text{Lifting}}^{3/4} + C_6 \cdot \mathbf{1}_{B'}, \quad B' := \{S : M(S) \leq M_0\}.$$

Meyn–Tweedie application. The set B' is bounded (since $M \leq M_0$ controls $\|\delta\|_\infty$), hence petite by Proposition 2. The chain is ψ -irreducible by Steps 1–2 of Theorem 1. Therefore [4, Thm. 14.3.7] applies: for $f_{\text{MT}} := \frac{\varepsilon_F}{2n^{3/2}} V_{\text{Lifting}}^{3/4}$, $\mathbb{E}_{\pi^*}[f] < \infty$, i.e. $\mathbb{E}_{\pi^*}[\Phi^{3/2}] < \infty$. Since $\Phi \leq \Phi^{3/2} + 1$, $\mathbb{E}_{\pi^*}[\Phi] < \infty$. $\square$

3.5 Doeblin minorization

Lemma 6 (Pushforward density). *Let $\xi_1, \dots, \xi_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, b)$ and $\Lambda(\xi) := (\xi_2 - \xi_1, \dots, \xi_n - \xi_1)$. The pushforward density ρ_Λ satisfies $\rho_\Lambda(\mathbf{y}) \geq b^{1-n}/2$ on $\mathcal{O} := \{\max_i |y_i| < b/4\}$.*

아이디어 스케치. fibre $\Lambda^{-1}(\mathbf{y}) \cap [0, b]^n$ 의 길이가 $\mathcal{O}$ 위에서 $> b/2$ 임을 직접 계산하면 균등밀도의 하한 $b^{1-n}/2$ 가 따라온다.

Proposition 2 (Doeblin minorization). *For any bounded set $\mathcal{B}_\Phi \subset \mathcal{X}^*$, there exist $\eta > 0$, $K^* < \infty$, and a probability measure ν such that $P^{K^*}(s, B) \geq \eta \nu(B)$ for every $s \in \mathcal{B}_\Phi$ and Borel B .*

아이디어 스케치. 네 가지 부품을 결합한다.

- (1) 길이 $K := n(T_{\max} + 1)$ 의 원도 안에는 block-flag 사건이 적어도 n 번 일어난다 (한 round 의 길이가 $\leq T_{\max} + 1$ 이므로).
- (2) 그 처음 n 번의 block-flag 가 $v_1, \dots, v_n$ 순서로 떨어지는 사건 $\mathcal{A}_1$ 의 확률은 $\geq \mu_{\min}^n > 0$ (μ_0 -i.i.d., height-state 와 독립).
- (3) flow-step 의 b'_s 를 $[0, \varepsilon_{\text{Doeb}}]$ ($\varepsilon_{\text{Doeb}} := b/(8K)$) 로 제약하는 사건 $\mathcal{A}_2$ 를 순차적으로 부과한다. block / non-block 의 구분이 $\mathcal{F}_{s-1}$ -가측이고 $b'_s \perp \mathcal{F}_{s-1}$ 이므로 block 증분의 marginal $\text{Unif}(0, b)$ 은 보존되며, $\mathcal{A}_2$ 의 확률은 $\geq (\varepsilon_{\text{Doeb}}/b)^{nT_{\max}} > 0$.
- (4) $\mathcal{A}_1 \cap \mathcal{A}_2$ 아래에서 한 원도가 만드는 δ -shift 는 Lemma 6 에 의해 박스 $\mathcal{O}$ 위에서 균등한 밀도 하한을 갖는다. $L := \lceil 16D_\Phi/b + 1 \rceil$ 개의 독립 원도를 합성하면 $\mathcal{B}_\Phi$ 전체를 cover; $K^* := LK$, η 는 위 확률들의 곱과 밀도 하한의 곱으로 strictly positive.

3.6 Theorem A: statement and proof

Theorem 1 (Balanced-regime convergence). *Assume the standing setting (connected $\mathcal{G}$, full-support μ_0 with $\mu_{\min} > 0$) and the two regime-defining hypotheses:*

- (A1) (**Balanced**) $\rho_i = 1/n$ for all $i \in \{1, \dots, n\}$;

(A2) (**Foster drift**) $(\mathcal{G}, \mu_0, b, T_{\max})$ satisfies Assumption 1: $B(S) \leq -\varepsilon_B M(S) + C_B$ for some $\varepsilon_B > 0$, $C_B \geq 0$.

Then there exists a deterministic matrix $C = [c_{ij}]_{i,j \in V}$ with $c_{ij} = c_{ij}(\mathcal{G}, b, T_{\max}) \geq 0$ such that

$$\frac{SD_{ij}^2(t)}{t} \xrightarrow{t \rightarrow \infty} c_{ij} \quad \text{a.s.},$$

and the limit c_{ij} is independent of the initial signal $\mathbf{y}$.

아이디어 스케치. Augmented state $S_t := (\delta_t, X_t, Z_t) \in \mathcal{X}^* := \mathbb{R}^{n-1} \times V \times \{0, \dots, T_{\max}\}$ ($\delta_{i,t} := h(v_i, t) - h(v_1, t)$) 의 round-skeleton $\{S_{t_r}\}$ 에서 시작.

(i) **Round-skeleton Foster–Lyapunov drift.** Lyapunov 함수 Φ 위에서

$$\mathbb{E}[\Phi(t_{r+1}) - \Phi(t_r) \mid S_{t_r}] \leq -\varepsilon_F M(S_{t_r}) + C_F \quad (\text{Fact 4}),$$

이며 $M > C_F/\varepsilon_F$ 밖에서 strictly negative. Foster–Lyapunov 정리 [4, Thm. 11.0.1] 가 $\{S_{t_r}\}$ 의 positive Harris recurrence 와 유일한 invariant probability π^* 의 존재를 보장한다.

(ii) **y-independence.** Proposition 2 의 Doeblin minorization 이 ν -irreducibility 를 주며 π^* 가 단일 closed Harris class 위에서 유일. ν 가 δ_0 에 무관하므로 π^* 도 초기 신호 $\mathbf{y}$ 에 무관.

(iii) **유한 모멘트 (Lifting argument).** $V_{\text{Lifting}} := \Phi^2$ 에 대해 one-step bound 와 Fact 3 의 Pigeon-hole ($\Phi \geq M^2/2$, $\Phi \leq nM^2$) 을 결합하면

$$\mathbb{E}[\Delta V_{\text{Lifting}} \mid S_{t_r}] \leq -\frac{\varepsilon_F}{2n^{3/2}} V_{\text{Lifting}}^{3/4} + C_6 \mathbf{1}_{\{M \leq M_0\}}$$

형태의 drift 가 나온다. [4, Thm. 14.0.1] 가 $\mathbb{E}_{\pi^*}[\Phi^{3/2}] < \infty$ 를 주고, 따라서 $\mathbb{E}_{\pi^*}[(\delta_i - \delta_j)^2] \leq 2\mathbb{E}_{\pi^*}[\Phi] < \infty$ (Proposition 1).

(iv) **Skeleton $\rightarrow$ 전체 chain 으로 transfer.** round 길이 $m_r \leq T_{\max} + 1$ 이 균등 유계이므로 inter-skeleton visit rate 는 skeleton 대비 최대 $T_{\max} + 1$ 배. Formal transfer 는 [4, Thm. 17.3.2].

(v) **Birkhoff LLN.** $g(S) := (\delta_i - \delta_j)^2$ 가 π^* -integrable 이므로 positive Harris chain 의 LLN [4, Thm. 17.1.7] 적용:

$$\frac{1}{t} \sum_{s=0}^t g(S_s) \xrightarrow{t \rightarrow \infty} \mathbb{E}_{\pi^*}[g] =: c_{ij} \quad \text{a.s.}$$

$g(S_s) = (\delta_{i,s} - \delta_{j,s})^2 = (h(v_i, s) - h(v_j, s))^2$ (공통항 $h(v_1, s)$ 가 상쇄) 이므로 좌변 = $SD_{ij}^2(t)/t$. $\square$

4 Theorem B

The proof of Theorem B requires only one external analytic tool: a classical lemma of Toeplitz that converts pointwise convergence of a sequence into the convergence of its weighted averages. We state it first as a wrapper lemma; the proof of Theorem B then maps the partial sums of D_s^2/s onto the Toeplitz array.

Lemma 7 (Toeplitz lemma). *Let $\{a_{ns}\}_{n,s \geq 1} \subset \mathbb{R}$ satisfy:*

- (T1) *There exists $c < \infty$ such that $\sum_{s=1}^{\infty} |a_{ns}| \leq c$ for all $n \geq 1$;*
- (T2) *$\sum_{s=1}^{\infty} a_{ns} \rightarrow 1$ as $n \rightarrow \infty$;*
- (T3) *for each fixed $s \geq 1$, $a_{ns} \rightarrow 0$ as $n \rightarrow \infty$.*

Then for any convergent sequence $x_s \rightarrow x \in \mathbb{R}$,

$$y_n := \sum_{s=1}^{\infty} a_{ns} x_s \xrightarrow{n \rightarrow \infty} x.$$

([6, §43]; see also [7, §3.2] and the original paper [5]. Kronecker's lemma is a corollary.)

Remark 1 (Notation conflict and rename). *The Toeplitz array in Lemma 7 is doubly indexed (denoted a_{ns}), but in the proof of Theorem B below the symbol a_s denotes a singly-indexed sequence ($a_s := D_s^2/s$). To avoid clash, when applying Lemma 7 in the proof below we rename the Toeplitz array to w_{ts} .*

Theorem 2 (Intermediate-regime convergence). *Assume the standing setting (connected $\mathcal{G}$, full-support μ_0) and the regime-defining hypothesis for the pair (i, j) :*

- (A3) (**Diffusive scaling**) ρ_i, ρ_j exist a.s., and the height-difference process $D_t := h(v_i, t) - h(v_j, t)$ satisfies

$$\frac{D_t^2}{t} \xrightarrow{t \rightarrow \infty} \sigma_{ij}^2 \in (0, \infty) \quad \text{a.s.} \quad (2)$$

(Note: (A3) implies $\rho_i = \rho_j$ a.s., since $\rho_i \neq \rho_j$ would give $D_t^2/t \rightarrow \infty$.)

Then

$$\frac{SD_{ij}^2(t)}{t^2} \xrightarrow{t \rightarrow \infty} \frac{\sigma_{ij}^2}{2} \quad \text{a.s.}$$

Proof. Since $\rho_i = \rho_j$ under (A3), the drift component of the height difference vanishes: writing $D_s = \bar{b}(\rho_i - \rho_j)s + \xi(s)$ from the decomposition in the proof of Theorem 3 below, the first term is identically zero, so $D_s = \xi(s)$ for all s .

Define $a_s := D_s^2/s$ for $s \geq 1$. By assumption (A3), $a_s \rightarrow \sigma_{ij}^2$ a.s. Decompose:

$$\begin{aligned} \frac{SD_{ij}^2(t)}{t^2} &= \frac{1}{t^2} \sum_{s=0}^t D_s^2 && (\because \text{def. of } SD_{ij}^2) \\ &= \frac{D_0^2}{t^2} + \frac{1}{t^2} \sum_{s=1}^t s a_s && (\because D_s^2 = s a_s \text{ for } s \geq 1) \\ &= \frac{D_0^2}{t^2} + \frac{t(t+1)/2}{t^2} \cdot \frac{\sum_{s=1}^t s a_s}{\sum_{s=1}^t s} && (\because \sum_{s=1}^t s = t(t+1)/2). \end{aligned}$$

Claim. $\frac{\sum_{s=1}^t s a_s}{\sum_{s=1}^t s} \xrightarrow{t \rightarrow \infty} \sigma_{ij}^2$ a.s.

Proof of Claim. Apply Lemma 7 with (cf. Remark 1)

$$w_{ts} := \begin{cases} \frac{s}{t(t+1)/2} & (s \leq t), \\ 0 & (s > t), \end{cases} \quad x_s := a_s = D_s^2/s.$$

Verifying the three Toeplitz conditions:

- **(T1)** $\sum_{s=1}^{\infty} |w_{ts}| = \frac{1}{t(t+1)/2} \sum_{s=1}^t s = 1$, so $c := 1$ works.
- **(T2)** $\sum_{s=1}^{\infty} w_{ts} = 1$ for every t , hence $\rightarrow 1$.
- **(T3)** For each fixed s , $w_{ts} = 2s/[t(t+1)] \rightarrow 0$ as $t \rightarrow \infty$.

Hence Lemma 7 applies with $x = \sigma_{ij}^2$, giving the Claim. $\square$

Substituting the Claim into the decomposition: $D_0^2/t^2 \rightarrow 0$, $t(t+1)/(2t^2) \rightarrow 1/2$, so

$$\frac{SD_{ij}^2(t)}{t^2} \xrightarrow{t \rightarrow \infty} 0 + \frac{1}{2} \cdot \sigma_{ij}^2 = \frac{\sigma_{ij}^2}{2} \quad \text{a.s.} \quad \square$$

Remark 2 (When does (2) hold?). *The condition $D_t^2/t \rightarrow \sigma^2 > 0$ means the height difference behaves diffusively ($|D_t| \sim \sigma\sqrt{t}$), as opposed to stationarily ($D_t = O(1)$, balanced) or ballistically ($D_t \sim \gamma t$, drift). This occurs when:*

- $\rho_i = \rho_j$ (drift cancels), but
- the augmented chain S_t is not positive Harris recurrent (global imbalance prevents mean-reversion of heights).

Concretely, on the Helm graph with degree-proportional μ_0 : pendant leaves L_i, L_j have $\rho_{L_i} = \rho_{L_j}$ (by symmetry), but the hub's linearly growing height propagates non-stationary noise through the flow. Since L_i and L_j are not directly connected, the noise arriving at each leaf is independent, making $D_t = h(L_i, t) - h(L_j, t)$ a random walk with step variance $\sim (b^2/3) \cdot 2\rho$, giving $\sigma^2 = 2\rho b^2/3$.

Remark 3 (Theorem B requires no recurrence machinery). *The proof reduces the convergence of SD_{ij}^2/t^2 to a single application of the Toeplitz lemma (Lemma 7)—no Foster–Lyapunov drift, no Doeblin minorization, no positive Harris recurrence is invoked. The hypothesis (2) of (A3) absorbs all the probabilistic content: it asserts that the height difference grows diffusively. Verifying (2) for specific graphs (e.g., the Helm graph above) requires analyzing the correlation structure of the walker, which we do empirically in the synthesis section.*

5 Theorem C: Drift regime

Theorem 3 (Drift-regime convergence). *Assume the standing setting (connected $\mathcal{G}$, full-support μ_0) and the regime-defining hypothesis for the pair (i, j) :*

(A4) (**Drift**) ρ_i and ρ_j exist a.s. with $\rho_i \neq \rho_j$.

Then

$$\frac{SD_{ij}^2(t)}{t^3} \xrightarrow{t \rightarrow \infty} \frac{\bar{b}^2(\rho_i - \rho_j)^2}{3} \quad \text{a.s.},$$

where $\bar{b} := \mathbb{E}[b'] = b/2$. The limit is strictly positive under (A_4) .

아이디어 스케치.

- (i) **Height 의 drift + martingale 분해.** 알고리즘 순서상 X_s 가 b'_s 추천 이전 에 결정되므로 $b'_s \perp (\mathcal{F}_{s-1}, X_s)$. 누적 height 를

$$h(v_i, t) = h(v_i, 0) + \bar{b} \sum_{s=1}^t \mathbf{1}\{X_s = v_i\} + M_t^{(i)}, \quad M_t^{(i)} := \sum_{s=1}^t (b'_s - \bar{b}) \mathbf{1}\{X_s = v_i\}$$

로 분리하면, 첫 항은 ρ_i 정의에 의해 $\bar{b}\rho_i t + o(t)$, 둘째 $M_t^{(i)}$ 는 conditional mean 0, 증분 $|\Delta M| \leq b/2$ 인 martingale 이고 martingale SLLN 에 의해 $M_t^{(i)}/t \rightarrow 0$ a.s. 결과적으로 $h(v_i, t) = \bar{b}\rho_i t + o(t)$.

- (ii) **Height difference asymptotics.** $\gamma := \bar{b}(\rho_i - \rho_j)$ 로 놓으면 $D(s) := h(v_i, s) - h(v_j, s) = \gamma s + \xi(s)$, $\xi(s) = o(s)$ a.s.

- (iii) **SD^2 전개와 선두항 추출.** 제곱전개하면

$$SD_{ij}^2(t) = \sum_{s=0}^t D(s)^2 = \gamma^2 \sum_{s=0}^t s^2 + 2\gamma \sum_{s=0}^t s \xi(s) + \sum_{s=0}^t \xi(s)^2.$$

선두항: $\sum_{s=0}^t s^2 = t^3/3 + O(t^2)$ 로 $\gamma^2 t^3/3 + O(t^2)$. 교차항 · 잔차항: $\xi(s)/s \rightarrow 0$ a.s. 로부터 Cesàro 형 평가 ($s \geq S_\epsilon$ 부터 $|\xi(s)| \leq \epsilon s$ 로 자르기) 로 모두 $o(t^3)$.

- (iv) **결론.** t^3 으로 나누면

$$\frac{SD_{ij}^2(t)}{t^3} \xrightarrow{t \rightarrow \infty} \frac{\gamma^2}{3} = \frac{\bar{b}^2(\rho_i - \rho_j)^2}{3} \quad \text{a.s.} \quad \square$$

Remark 4 (Existence of ρ_i). The assumption that ρ_i exists a.s. is verified empirically in all examples below: the running average $\frac{1}{t} \sum_{s=1}^t \mathbf{1}\{X_s = v_i\}$ stabilizes well before the SD^2/t^3 ratio does. A general sufficient condition is a strong law of large numbers for the empirical occupation measures of the deposition process; in the balanced regime, this follows from the Harris recurrence established in Theorem 1. In the drift regime, a rigorous proof of ρ_i -existence for general graphs remains open, though the weak ergodicity of the walker process [1, Thm. A.5] provides supporting evidence.

Remark 5 (No recurrence machinery needed). Theorem 3 requires neither Foster–Lyapunov nor Doeblin minorization. The convergence is a direct consequence of the strong law and the Toeplitz lemma (weighted Cesàro mean with weights s^2). The difficulty of the balanced regime (Theorem 1) lies precisely in the absence of the dominant $O(t)$ drift: the $o(t)$ fluctuations $\xi(s)$ must themselves be controlled by recurrence machinery to show $\sum \xi(s)^2/t \rightarrow c_{ij}$.

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